Background-Disjoint Cross-Validation Reveals Hidden Failure Modes in CNN Gravitational Lens Classifiers

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ABSTRACT

CNN-based gravitational lens finders are widely used in survey pipelines and are commonly trained using simulated arc injections into real galaxy images. Injection pipelines frequently generate multiple training samples from the same background galaxy. Under standard image-level cross-validation, images derived from a common background can appear in both training and test folds, creating a leakage pathway through which models can memorise background morphology and achieve optimistic performance estimates.

Background-disjoint cross-validation (BDCV) eliminates this leakage channel by assigning all images derived from a given background galaxy to a single fold. Applied to a controlled dataset of 2,200 single-band images (200 arc injections and 2,000 negatives) drawn from 217 unique backgrounds, a convolutional neural network with approximately 3.6×10^5 parameters yields FPR = 0.106 and AUC = 0.968 under standard five-fold cross-validation. Background-disjoint evaluation yields FPR = 0.145 and AUC = 0.917, revealing a 37 per cent increase in false positives and a substantial reduction in recovery.

False positives are extremely concentrated across background galaxies. The per-background false-positive distribution has Gini coefficient $G = 0.879$, exceeding the maximum value obtained under 2,000 permutation replicates (0.530) by a margin of 0.35 and placing the observed statistic completely outside the null support. Classical photometric and morphological descriptors fail to distinguish catastrophic from clean backgrounds, yet separation emerges in the learned representation (embedding-space AUC = 0.947). Replication on an independent parametric simulation of 150 Sérsic-profile backgrounds yields standard CV FPR = 0.107 and BDCV FPR = 0.180.

Image-level cross-validation can therefore substantially underestimate false-positive burden in injection-trained lens classifiers. Background-disjoint cross-validation provides a simple evaluation safeguard capable of revealing background-dependent shortcut behaviour.

Key words: gravitational lensing: strong – methods: statistical – techniques: image processing – surveys

1 INTRODUCTION

Gravitational strong lensing — the formation of arcs and Einstein rings when a massive foreground galaxy deflects light from a background source — provides powerful constraints on dark matter distributions, cosmological parameters, and the mass structure of galaxies. Wide-field photometric surveys are expected to yield tens of thousands of strong lens candidates, far exceeding the capacity for visual inspection.

CNN-based lens finders are now widely used in survey pipelines (**???**). The standard training approach injects simulated gravitational arcs into real galaxy images and trains a binary classifier to distinguish injected images from unmodified backgrounds.

Dataset bias and shortcut learning represent well-documented challenges in machine-learning systems. Models frequently exploit correlations present in training data rather than causal signals relevant to the target task (**??**). Similar concerns arise in evaluation methodology, where standard validation protocols can mask distribution shifts between training and deployment environments (**?**). Injection-trained lens classifiers possess a structural property that creates an additional vulnerability: multiple training samples may originate from the same background galaxy. Under image-level cross-validation, this structure allows background morphology to appear simultaneously in training and evaluation partitions, potentially inflating apparent performance. To date, background identity has not been used as a constraint in cross-validation protocols for injection-trained lens classifiers. Because injection pipelines are widely used to train automated lens finders for current and upcoming surveys, evaluation protocols that overlook this dependency structure may systematically underestimate the operational false-positive burden.

This paper introduces background-disjoint cross-validation (BDCV) and demonstrates that this methodological change reveals substantial background-dependent failure modes invisible to standard evaluation. Group-wise cross-validation has been applied in other domains, for example patient-

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level splits in medical imaging. The contribution here is the demonstration that background-identity constraints reveal a systematic and quantifiable failure mode in injection-trained astronomical classifiers.

2 DATA AND METHODS

2.1 Dataset

The dataset consists of 2,200 single-band 101×101-pixel images drawn from 217 unique background galaxies: 200 positives (simulated arc injections) and 2,000 negatives (unmodified backgrounds). Each background contributes approximately 9–10 negatives and 0–1 positives. Images are sky-subtracted single-band cutouts.

2.2 Model

The classifier has approximately 3.6×10^5 parameters with a 128-dimensional penultimate embedding layer, using convolutional blocks with batch normalisation and ReLU activations, global average pooling, and a fully connected classification head.

2.3 Background-disjoint cross-validation

BDCV treats the background galaxy as the unit of assignment: all images derived from a given background are assigned to the same fold. With 5-fold BDCV on 217 backgrounds, each test fold contains images from approximately 43–44 unique backgrounds unseen during training. Implementation requires only that images be grouped by background identifier during fold construction (e.g. GroupKFold in scikit-learn) and therefore imposes no additional computational cost relative to standard cross-validation.

2.4 Evaluation metrics

Performance is summarised by pooled false-positive rate (FPR), true-positive rate (TPR), and mean fold-level AUC. The FPR target of 0.10 follows conventions used in the lens-finding literature; the primary result, concentration of errors across backgrounds, is independent of this threshold. A background is classified as catastrophic when FPR = 1.0 across all negatives in that group. False-positive concentration is quantified using the Gini coefficient computed over per-background FP counts, with significance assessed by permutation test (2,000 replicates, seed = 42). The permutation test preserves the total number of false positives while randomly reallocating those errors across background groups, representing the concentration expected under background-independent errors.

3 RESULTS

Background-disjoint evaluation reveals two striking empirical properties of the classifier. First, standard image-level cross-validation substantially underestimates the false-positive burden when background identity is ignored during fold construction. Second, false-positive errors are highly concentrated across background galaxies rather than approximately uniform. Aggregating fold-level predictions yields 289 false positives arising from only 32 of 217 background galaxies; 185 backgrounds produce no errors. The false-positive distribution has Gini coefficient $G = 0.879$, far outside a permutation null generated by random reallocation of errors across backgrounds (null mean 0.463, maximum 0.530, 2,000 permutations). The observed statistic exceeds the entire null support by a margin of 0.35, indicating that the concentration cannot plausibly arise from background-independent error allocation. Subsequent subsections quantify the evaluation discrepancy, examine diagnostic evidence for the failure mode structure, and present an independent replication.

Table 1. Standard versus background-disjoint cross-validation. Same dataset, architecture, hyperparameters, and seed; protocols differ only in fold construction. Standard CV yields an apparently acceptable FPR (0.106) but BDCV reveals a 37 per cent higher error rate and markedly lower recovery.

Evaluation protocol	AUC	TPR	FPR
Standard KFold	0.968	0.935	0.106
Background-disjoint KFold	0.917	0.781	0.145

3.1 Standard versus background-disjoint cross-validation

Standard and background-disjoint 5-fold cross-validation results are compared in Table **??**. Both protocols used the same dataset, architecture, hyperparameters, and random seed, differing only in fold construction. Performance differs substantially. Under ordinary random 5-fold cross-validation the pooled FPR is 0.106, close to the nominal target of 0.10, and the pooled TPR is 0.935. Under BDCV the pooled FPR increases to 0.145 (+37 per cent), while TPR decreases to 0.781 (−15.4 percentage points). Mean validation AUC declines from 0.968 ± 0.028 to 0.917 ± 0.087.

The change in performance is accompanied by a marked increase in error heterogeneity. Standard CV identifies 9 catastrophic background groups (those on which every negative is misclassified), whereas BDCV reveals 29 — a 3.2-fold increase. Fold-to-fold FPR standard deviation increases 2.3× and TPR standard deviation increases 4.1× under BDCV. Twenty backgrounds that appear completely clean under standard evaluation produce FPR = 1.0 when the model cannot memorise them.

These results indicate that standard image-level cross-validation can produce optimistic performance estimates when background realisations appear in both training and test partitions. The injected-data construction is critical: because arc injections are generated independently of background morphology, no astrophysically legitimate correlation exists between background structure and lens presence in this dataset; any predictive dependence on background morphology therefore reflects shortcut use within the learned representation rather than genuine lensing signal.

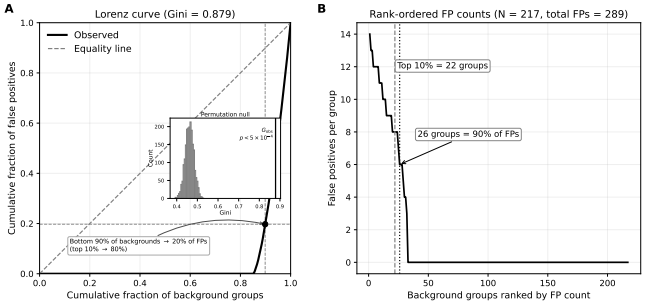


Figure 1. Left: false positives per background group, ranked in decreasing order. Right: Lorenz curve across all 217 groups. Gini = 0.879; 22 groups (10.1 per cent) account for 80 per cent of all 289 errors. Permutation $p < 5 \times 10^{-4}$.

3.2 False-positive concentration across background galaxies

Under BDCV the pooled FPR is 0.145, exceeding the target of 0.10. This excess is not distributed uniformly. Aggregating from the five fold-level prediction files yields 289 false positives across 217 backgrounds (mean = 1.33 per group). Of 217 backgrounds, 185 produce no false positives. The remaining 32 backgrounds produce at least one false positive, while 29 backgrounds are catastrophic (FPR = 1.0). Each catastrophic background contributes approximately 9–10 negatives; FPR = 1.0 therefore represents consistent misclassification of all images from that background rather than a single-image artifact. Error concentration is itself highly skewed: 22 backgrounds (10.1 per cent) account for 80 per cent of all errors (Fig. **??**). The Gini coefficient is $G = 0.879$; a permutation test yields $p < 5 \times 10^{-4}$ (2,000 permutations). The observed statistic exceeds the maximum value obtained in 2,000 permutations by 0.35, placing the observed concentration far outside the range expected under background-independent error allocation.

Additional perturbation tests produced qualitatively consistent behaviour but introduced masking artefacts that complicate interpretation; the primary inference therefore relies on the concentration statistic and representation analysis.

Visual inspection of the six highest-error catastrophic groups confirms that the operative confounders are large-scale morphological features of the background galaxies themselves (Fig. **??**). Confounders include tidal streams, off-axis compact source configurations, and absorbing features with arc-like curvature. The injected arc signal is not visible at the display stretch used, confirming that the classifier response is dominated by pre-existing background structure.

Kolmogorov–Smirnov tests on sky coordinates of catastrophic versus clean groups yield $p = 0.81$ (RA) and $p = 0.68$ (Dec), ruling out spatial clustering or field-level artefacts (Fig. **??**).

Across three rounds of progressively detailed morphological analysis, no tested interpretable morphology statistic distinguished catastrophic from clean groups. Spearman correlations between CNN logit and scalar metrics (peak surface brightness, total flux, half-light radius, concentration index) across all 2,000 negatives are all $|\rho| < 0.012$, $p > 0.6$ (Fig. **??**). Peak surface brightness, the classical predictor most likely to correlate with both background morphology and classifier response, shows Spearman $|\rho| < 0.02$ ($p > 0.6$), ruling out brightness as the operative confounder. Azimuthally averaged radial profiles show no significant difference at any of 35 radius bins. A panel of 15 non-radial 2-D features — Sobel edge strength and orientation entropy, asymmetry residuals, ellipticity, arcness, and PCA components — yields zero of 30 feature–aggregation pairs at $p < 0.05$, with all effect sizes $|r| < 0.2$. The operative feature therefore likely resides in a distributed representation that is not preserved by the summary statistics considered here.

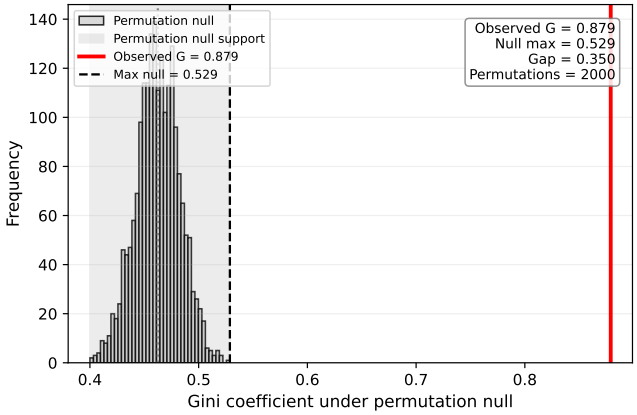


Figure 2. Permutation null distribution of the false-positive Gini coefficient (2,000 replicates; FP counts randomly reallocated across 217 background groups preserving the observed total of 289). The null distribution (grey histogram) is compact and approximately symmetric (mean = 0.463, SD = 0.020, max = 0.530). The observed Gini coefficient $G = 0.879$ (red vertical line) lies 0.35 above the null maximum, exceeding all 2,000 replicates ($p < 5 \times 10^{-4}$). The empty gap between the null support and the observed value indicates that the concentration cannot plausibly arise from random error allocation across backgrounds.

3.3 Embedding-space separation and Grad-CAM attribution

The following analyses investigate the representation-level structure of the failure mode and are presented as mechanistic diagnostic evidence rather than as the primary statistical result.

Catastrophic and clean background groups were not distinguishable using the classical photometric and morphology descriptors tested above. In contrast, the same groups are strongly separable in the model's learned representation. Using the 128-dimensional penultimate embedding extracted from out-of-fold models, a leave-one-out logistic regression classifier distinguished catastrophic (FPR = 1.0) from clean (FPR = 0.0) background groups with AUC = 0.947 (permutation $p < 0.005$). Catastrophic groups do not form a visually distinct cluster (silhouette = 0.036); nearest-neighbour purity is modest ($k = 5$: 0.64; $k = 10$: 0.56). Discriminative signal is carried by minor PCA components (PC3, $p = 0.001$; PC5, $p = 3.6 \times 10^{-5}$) rather than dominant variance directions. When coloured by mean classifier logit, the embedding reveals a continuous mapping from background type to prediction score, with catastrophic groups occupying regions of

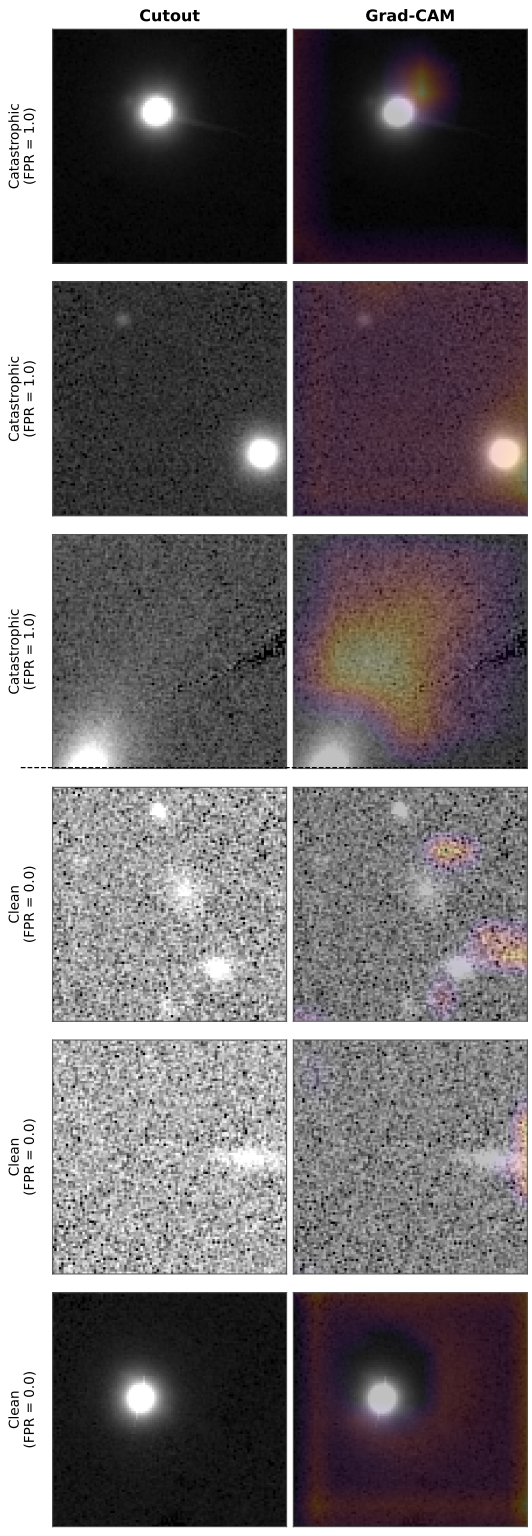


Figure 3. Catastrophic background groups. Each row shows (left) the background galaxy, (centre) a false-positive negative, and (right) a simulated arc injection from the same background at identical display stretch. The arc signal is not visible; classifier activation is driven by pre-existing background morphology.



Figure 4. Sky distribution of catastrophic (red crosses) and clean (blue circles) background groups. KS tests: $p = 0.81$ (RA), $p = 0.68$ (Dec). No spatial clustering.

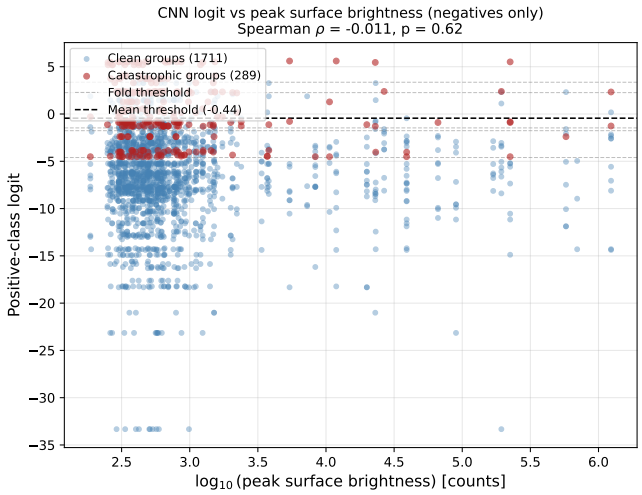


Figure 5. CNN logit vs. $\log_{10}$ peak surface brightness for all 2,000 negatives. Catastrophic (red) and clean (blue) images overlap across the full brightness range. Spearman $|\rho| < 0.02$, $p > 0.6$ for all scalar metrics.

feature space associated with systematically higher positive-class logits (Fig. **??**).

This result establishes that the failure mode is encoded in the model's internal representation even though it is not captured by the tested classical descriptors. Within the feature families evaluated here, no standard astronomical summary statistic reproduces this separation.

To localise the spatial basis of this learned failure mode,

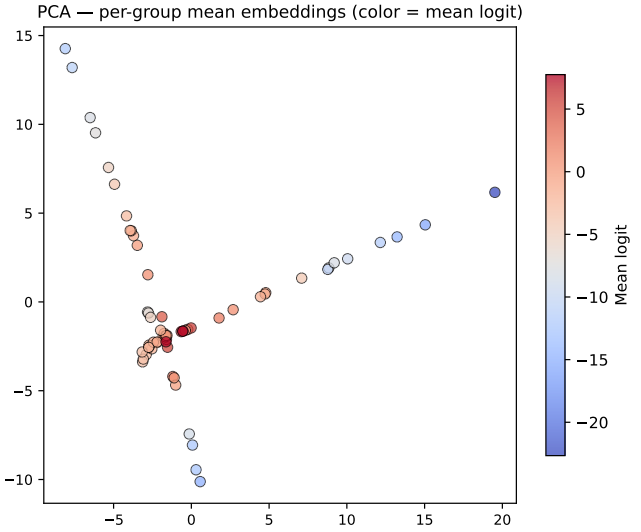


Figure 6. PCA of per-group mean embeddings coloured by mean positive-class logit. Catastrophic groups (warm colours, high logit) concentrate centrally; clean groups (cool colours) scatter to the periphery. LOO logistic AUC = 0.947 ($p < 0.005$) in the full 128-dimensional space.

Grad-CAM[1] attribution maps (**?**) were computed for catastrophic and clean background groups. Qualitative examples are shown in Fig. **??**. Catastrophic backgrounds exhibit stronger and more spatially coherent positive-class attribution on off-centre galaxy structure, including curved outer features and surrounding halo emission, whereas clean backgrounds typically show weaker or less coherent activation.

Quantitative attribution summaries were computed for all 29 catastrophic groups (289 images) and 32 matched clean groups (313 images). Attribution concentration was not stable across two independent retraining runs ($p = 2.3 \times 10^{-6}$ in one run, $p = 0.068$ in another) and is not treated as a primary inferential result. Two attribution patterns were robust. Peak outer attribution is consistently higher for catastrophic groups (0.454 ± 0.426 vs 0.341 ± 0.397, $p = 2.2 \times 10^{-4}$), confirming activation in the 10–35 pixel annulus where lensing arcs appear in training. Attribution asymmetry is consistently lower for catastrophic groups (median 3.5 vs 10.0, $p < 10^{-11}$), indicating more spatially balanced activation (Fig. **??**).

The embedding and attribution results are complementary. The embedding analysis shows that catastrophic and clean backgrounds are strongly separable in the learned representation despite being indistinguishable under tested classical morphology summaries. The Grad-CAM analysis shows that this representation is expressed spatially through increased reliance on off-centre background structure at arc-relevant radii. Together these analyses are consistent with a

[1] Grad-CAM attribution maps were computed from models retrained under the same background-disjoint fold structure because fold-level checkpoints from the canonical experiment were not retained. Attribution analyses therefore characterise typical model behaviour under the protocol rather than the exact parameter realisations used in the primary evaluation.

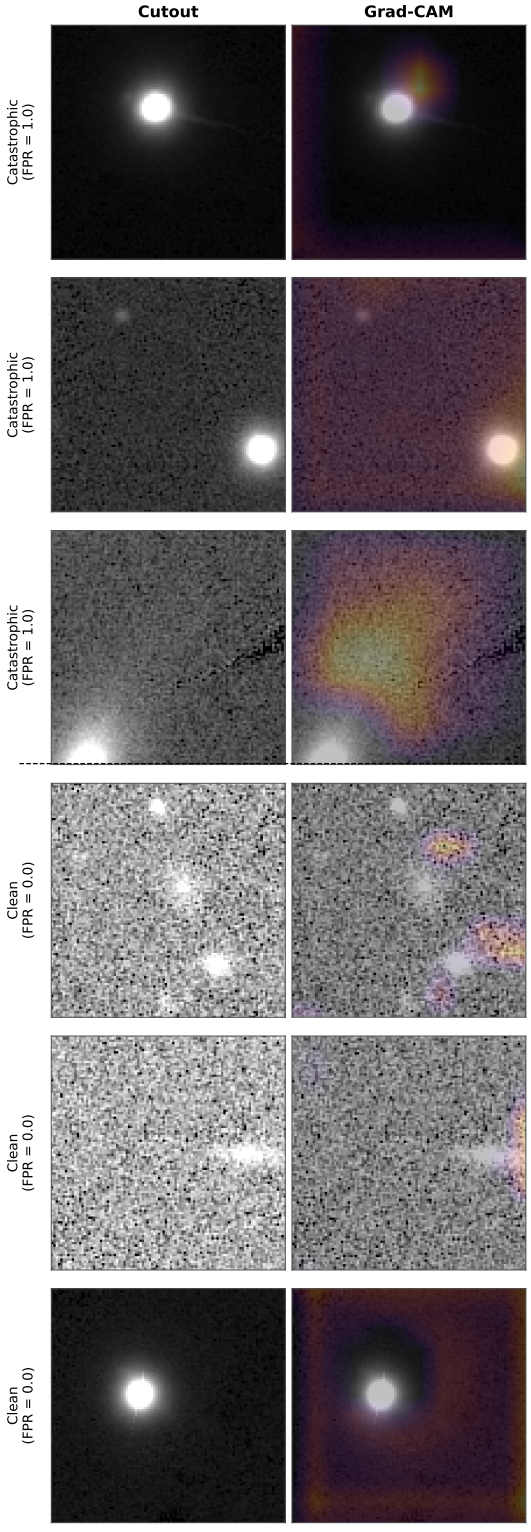


Figure 7. Grad-CAM attribution for catastrophic (top 3 rows, FPR = 1.0) and clean (bottom 3 rows, FPR = 0.0) background groups. Left: normalised cutouts; right: Grad-CAM overlays. Catastrophic groups show stronger, more coherent attribution on outer halo structure consistent with a morphology-driven shortcut in which curved halo structure activates the positive-class pathway.

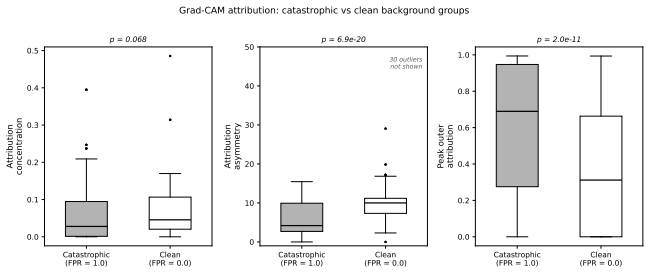


Figure 8. Quantitative Grad-CAM summary. Left: attribution concentration (not stable across retraining runs; descriptive only). Centre: attribution asymmetry (lower for catastrophic — more spatially balanced). Right: peak outer attribution (higher for catastrophic — outer halo activation). Asymmetry and peak outer attribution are stable across independent retraining runs and are treated as primary results.

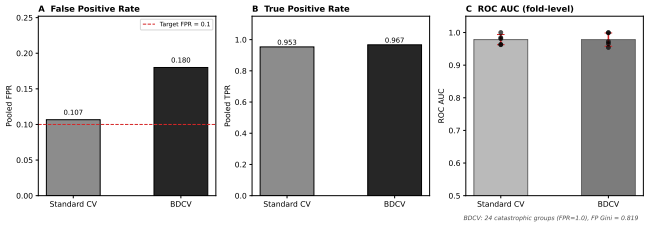


Figure 9. Replication on an independent parametric simulation (150 Sérsic-profile backgrounds, LENSTRONOMY pipeline). Panel A: pooled FPR under standard CV (0.107) and BDCV (0.180); dashed red line indicates the FPR target of 0.10. Panel B: pooled TPR. Panel C: fold-level AUC with individual fold values shown. The standard-CV vs. BDCV gap and elevated fold-level variance under BDCV are consistent with the primary PS1 result, supporting robustness across simulation settings. BDCV: 24 catastrophic groups (FPR = 1.0), FP Gini = 0.819.

background-dependent failure mode in which the CNN has learned a representation that maps specific galaxies to regions of feature space associated with elevated lens logits, expressed through activation on distributed outer-galaxy structure rather than any simple brightness or compactness cue.

3.4 Independent replication

To test whether the observed error concentration is specific to the PS1 empirical backgrounds used in the primary experiment, the analysis was repeated on an independent parametric simulation generated with the LENSTRONOMY pipeline (**?**). Replication on 150 Sérsic-profile backgrounds yields standard CV FPR = 0.107 and BDCV FPR = 0.180. The standard-CV vs. BDCV gap and elevated fold-level variance under BDCV are consistent with the primary PS1 result, supporting robustness across simulation settings. BDCV identifies 24 catastrophic groups (FPR = 1.0) and FP Gini = 0.819 (Fig. **??**).

Taken together, the concentration result, the spatial null result, the morphological null results, the Grad-CAM attribution analysis, and the embedding geometry provide a coherent account of the observed failure mode. The CNN has learned a background-dependent feature representation encoded in minor principal components of the 128-dimensional embedding space that are not captured by the standard photometric and morphology descriptors evaluated here. This representation drives classification decisions that are invisible to background-unaware evaluation: the same backgrounds that produce FPR = 1.0 under BDCV appear indistinguishable from safe backgrounds by every classical morphology metric, yet are strongly separated in the CNN's own internal representation. Background-disjoint cross-validation is a low-cost evaluation safeguard that can reveal background-dependent failure modes hidden by image-level cross-validation.

4 DISCUSSION

4.1 Mechanism of the shortcut

The injected-data design is critical for causal interpretation. Because simulated lens signals are assigned independently of background identity, no astrophysically legitimate correlation exists between background morphology and the presence of a lens in this dataset. Predictive dependence on background structure must therefore arise from shortcut use rather than from any true lensing signal. This pattern is characteristic of shortcut learning in deep convolutional models (**?**), where predictive signals arise from combinations of weak cues distributed across many spatial filters rather than from a single physically interpretable feature. The permutation null tests whether the observed error concentration exceeds what would arise if false positives were allocated independently of background identity; within-group error correlation is the phenomenon under investigation, not a confound of the null model.

If catastrophic backgrounds simply represented intrinsically difficult morphologies, performance would be expected to degrade gradually across backgrounds under both evaluation protocols. Instead, the failure is discrete and background-specific: several backgrounds produce FPR = 1.0 under BDCV while remaining error-free under standard cross-validation, whereas most backgrounds produce no errors under either protocol. This pattern is difficult to reconcile with a continuous difficulty gradient and is consistent with background-specific memorisation.

In real astronomical data, background morphology correlates with lensing probability through the host galaxy population. The present result is specific to injection-trained datasets where that correlation is absent by construction.

4.2 Implications for injection-trained lens finders

Standard cross-validation suggests the model satisfies the target false-positive rate, whereas background-disjoint evaluation reveals a 37 per cent higher FPR and substantially lower recovery, demonstrating that image-level splitting conceals background leakage. Because many lens-finding pipelines are trained using simulated arc injections into real galaxies, similar correlations between training and test samples can arise whenever evaluation protocols ignore background identity. Background-disjoint cross-validation provides a simple and computationally inexpensive evaluation safeguard that can

reveal such background-dependent failure modes before models are deployed in survey pipelines. The aggregate FPR increase from 0.106 to 0.145 does not fully convey the practical consequence of the failure mode. Standard evaluation identifies 9 catastrophic backgrounds; BDCV reveals 29. A classifier deployed after standard evaluation will systematically misclassify every image from 20 background galaxy types that its own evaluation protocol identified as safe.

4.3 Limitations

Several limitations constrain the scope of the results. The analysis is restricted to a single-band, injection-trained setting, and multiband information may reduce reliance on morphology-based shortcuts by providing colour contrast between foreground lenses and background sources. The experiment is conducted on a controlled dataset comprising 217 unique background galaxies and one CNN architecture; results should be interpreted as a controlled diagnostic demonstration. The sample is adequate for the concentration analysis, where the relevant unit is the background group rather than the individual image, but modest as a training corpus. The embedding-space separation result uses the same model whose failure is being characterised. The analysis demonstrates that the failure mode is encoded within the learned representation but cannot independently establish which specific property of background morphology is causally responsible. The observed Gini coefficient exceeds the permutation null maximum by 0.35, a margin that cannot be attributed to small-sample variance. Whether attention-based or transformer architectures are more or less susceptible to this failure mode is unknown; the concern is likely general to any injection-trained model, but calibration on other architectures remains warranted. Mitigation strategies such as adversarial invariance, targeted augmentation, or hard-negative retraining are not benchmarked here. The contribution is methodological and diagnostic: to show that background-disjoint evaluation can reveal a background-dependent failure mode that standard image-level validation may miss.

5 CONCLUSIONS

Background-disjoint cross-validation reveals substantial failure modes in CNN lens classifiers invisible to standard evaluation. The principal findings are:

(i) Standard CV achieves FPR = 0.106 and AUC = 0.968, apparently meeting the target. BDCV reveals FPR = 0.145 (+37 per cent) and AUC = 0.917.

(ii) Error distribution is extremely concentrated: Gini = 0.879, permutation $p < 5 \times 10^{-4}$; the observed statistic exceeds the permutation null maximum by 0.35; 22 of 217 backgrounds account for 80 per cent of all 289 errors. Standard CV identifies 9 catastrophic backgrounds; BDCV reveals 29.

(iii) Spatial KS tests ($p = 0.81$, 0.68) rule out field-level artefacts.

(iv) All tested morphological metrics — scalar, radial, and 15 non-radial 2-D features — fail to distinguish catastrophic from clean backgrounds.

(v) A linear classifier in the CNN's 128-dimensional penultimate embedding separates catastrophic from clean groups with LOO AUC = 0.947 ($p < 0.005$).

(vi) Background-disjoint evaluation provides a simple, low-cost safeguard that can reveal background-dependent failure modes before deployment.

(vii) Independent replication on 150 Sérsic-profile backgrounds (LENSTRONOMY pipeline) yields standard CV FPR = 0.107 and BDCV FPR = 0.180, with 24 catastrophic groups and FP Gini = 0.819, confirming that the result is not specific to the PS1 empirical backgrounds.

Background-disjoint cross-validation therefore provides a simple evaluation safeguard capable of revealing background-dependent shortcut behaviour that remains hidden under conventional image-level validation.

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